

Model-Aware Positive Stieltjes Quadrature: From Structural Minimax to Gauss–Legendre

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Abstract

Generalized Gaussian and rationally exact quadrature already exploit non-polynomial structure, while rational minimax approximation under interpolation constraints is an established approximation problem. We consider the more specific design of positive symmetric Stieltjes-response quadrature rules that minimize relative error uniformly over a continuum of Lorentzian widths while matching a variable prefix of polynomial moments. Within this class, n positive node pairs and moment depth k give the feasible range $0 \leq k \leq 2n$, an admissible dimension $2n - k$, and the Gauss–Legendre rule at $k = 2n$. For $k < 2n$, a difference-numerator degree bound yields a $2n - k + 1$ -point alternation certificate. High-precision hierarchies for $n = 2, 3, 4$ quantify the trade-off between structural accuracy and polynomial protection. For a smooth residual with Chebyshev coefficients $|a_j| \leq C\rho^{-j}$, an exact coefficient-space norm combines with the Lorentzian error to give the model-aware risk $G_k(\rho) + \tau S_k$. Conservative numerical enclosures assign a unique rule on 97.9067% of the tested parameter domain. Seventy-two multi-start reconstructions recover the upper-half rules, and a shifted-Lorentzian stress test supports local rather than universal robustness. The distinguishing contribution is the combined positive minimax path, exact selector, and validated decision map, not positivity, moment matching, or rational approximation separately.

Keywords: numerical quadrature, Stieltjes functions, rational minimax approximation, moment constraints, Gauss–Legendre quadrature, validated numerics

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1. Introduction

Polynomial exactness is a central organizing principle of numerical quadrature, but it is not the only way to spend a fixed node budget. When the dominant difficulty is a nearby pole, quadrature may instead incorporate rational functions chosen to match that structure [1]. Generalized Gaussian quadrature extends positive exactness to Chebyshev systems and is naturally described through moment-space geometry [2]. Positive interpolatory formulas and optimization-based moment-matching constructions provide additional general frameworks [6, 7, 8].

A second neighboring direction begins with rational approximation. Modern algorithms compute rational minimax approximants through adaptive barycentric representations and Remez-type iterations [3]. Interpolation-constrained rational minimax approximation, including higher-order constraints, is also established [4]. For Markov and Stieltjes functions, relative rational approximation has a substantial theory [9]. Most directly, quadrature formulas can be derived from rational approximations of Cauchy transforms, with poles and residues becoming nodes and weights and Gauss quadrature connected to Padé approximation at infinity [5].

These results rule out several broad novelty claims: pole-aware quadrature, positive exact quadrature, rational minimax, equioscillation, and constrained rational approximation are not new separately. The narrower problem studied here is to combine them in a model-aware design. At fixed node-pair count, we require a positive symmetric Stieltjes response, match a variable prefix of polynomial moments, and minimize relative error uniformly over a continuum of Lorentzian widths. The resulting family spans a structural minimax rule and the Gauss–Legendre endpoint. The practical question is then not only how to compute the rules, but how to select the moment depth when a smooth uncertain residual is superposed on the structural component.

Within this stated class, the paper makes the following contributions:

- (i) it formulates the positive variable-moment continuum-minimax problem and characterizes its feasible range, dimension, and Gauss–Legendre endpoint;
- (ii) it gives a short class-specific degree/zero-count argument leading to a $2n - k + 1$ -point alternation certificate for every nonzero-dimensional level;

- (iii) it computes the complete path for $n = 2, 3, 4$, including the upper-half levels between the earlier lower-half designs and the Gaussian endpoint;
- (iv) it derives an exact mixed structural/smooth risk and uses it to select the moment depth across a two-parameter model domain;
- (v) it supplies a conservative validated decision map, a shifted-pole scope test, and independent multi-start reconstruction.

The feasible-range theorem and the Gaussian endpoint are important structural components of the framework, but are not claimed as stand-alone inventions of moment or generalized Gaussian theory. A targeted literature search did not identify a direct source combining positivity, continuum Lorentzian minimax, variable moment depth, an exact mixed selector, and a validated parameter partition. This is not an exhaustive proof of novelty; specialist prior-art review remains necessary. The claims are restricted to the symmetric positive class and do not imply universal superiority over Gaussian, adaptive, or problem-specific quadrature.

2. Problem formulation

Let

$$I[f] = \int_{-1}^1 f(x) dx, \quad L_b(x) = \frac{1}{x^2 + b^2}, \quad b \in B = [0.08, 0.12].$$

With $s = b^2$ and $y = x^2$,

$$F(s) = I[L_{\sqrt{s}}] = \int_0^1 \frac{y^{-1/2}}{s + y} dy = \frac{2}{\sqrt{s}} \arctan \frac{1}{\sqrt{s}}.$$

A symmetric n -pair rule has the form

$$Q[f] = \sum_{j=1}^n w_j \{f(-x_j) + f(x_j)\}, \quad 0 < x_1 < \cdots < x_n < 1, \quad w_j > 0.$$

Writing $y_j = x_j^2$, the response on the structural family is the positive Stieltjes rational function

$$r_Q(s) = Q[L_{\sqrt{s}}] = 2 \sum_{j=1}^n \frac{w_j}{s + y_j}.$$

For an integer $k \geq 0$, define the moment-constrained class

$$\mathcal{A}_{n,k} = \left\{ Q : \sum_{j=1}^n w_j y_j^\ell = \frac{1}{2\ell+1}, \quad \ell = 0, \dots, k-1 \right\}.$$

For $k = 0$, no moment condition is imposed.

Lemma 1 (Equivalent interpretations of the constraints). *For $k \geq 1$, the following conditions are equivalent:*

- (a) $Q \in \mathcal{A}_{n,k}$;
- (b) Q is exact for every polynomial of degree at most $2k-1$;
- (c) $r_Q(s) - F(s) = O(s^{-k-1})$ as $s \rightarrow \infty$.

Proof. Symmetry gives exactness for odd polynomials. For even monomials,

$$Q[x^{2\ell}] = 2 \sum_{j=1}^n w_j y_j^\ell, \quad I[x^{2\ell}] = \frac{2}{2\ell+1},$$

which proves the equivalence of (a) and (b). The expansion

$$\frac{1}{s+y} = \sum_{\ell=0}^{m-1} (-1)^\ell y^\ell s^{-\ell-1} + O(s^{-m-1})$$

is uniform for $y \in [0, 1]$. Integrating it against $y^{-1/2} dy$ and applying it to the discrete measure $2 \sum_j w_j \delta_{y_j}$ proves the equivalence with (c). $\square$

3. Feasible-range characterization

Let

$$\mathcal{P}_n = \{(y, w) : 0 < y_1 < \dots < y_n < 1, w_j > 0\} \subset \mathbb{R}^{2n},$$

and define the moment map

$$M_k(y, w) = \left(\sum_{j=1}^n w_j y_j^\ell \right)_{\ell=0}^{k-1}.$$

Theorem 2 (Feasible range, dimension, and Gaussian endpoint). *Within the positive ordered class:*

- (i) $\mathcal{A}_{n,k}$ is nonempty for every $0 \leq k \leq 2n$;
- (ii) for $0 \leq k \leq 2n$, it is a smooth manifold of dimension

$$\dim \mathcal{A}_{n,k} = 2n - k;$$

- (iii) $\mathcal{A}_{n,k}$ is empty for $k > 2n$;
- (iv) $\mathcal{A}_{n,2n}$ consists of the positive half of the classical $2n$ -point Gauss–Legendre rule.

Proof. Nonemptiness follows because the $2n$ -point Gauss–Legendre rule satisfies all moment conditions through $k = 2n$, and hence every prefix.

To prove the rank statement, suppose a linear combination of the rows of DM_k vanishes and define

$$p(y) = \sum_{\ell=0}^{k-1} c_\ell y^\ell.$$

The columns obtained by differentiating with respect to w_j give $p(y_j) = 0$, while the columns obtained by differentiating with respect to y_j give $w_j p'(y_j) = 0$. Positivity implies $p'(y_j) = 0$. Thus every y_j is a double root. For $k \leq 2n$, however, $\deg p \leq 2n - 1$, so $p \equiv 0$. Therefore DM_k has full row rank k , and the implicit-function theorem gives dimension $2n - k$.

For impossibility, suppose moments through index $2n$ were matched. The polynomial

$$g(y) = \prod_{j=1}^n (y - y_j)^2$$

has degree $2n$, is nonnegative, and is not identically zero. The discrete transformed rule gives $2 \sum_j w_j g(y_j) = 0$, whereas

$$\int_0^1 g(y) y^{-1/2} dy > 0,$$

a contradiction.

At $k = 2n$, the transformed n -node rule is exact through degree $2n - 1$ for the measure $y^{-1/2} dy$. It is therefore its Gaussian quadrature rule. Under $y = x^2$, the symmetric rule is exact through degree $4n - 1$, and hence is the $2n$ -point Gauss–Legendre rule. $\square$

Remark 3. The previously studied range $0 \leq k \leq n$ is the lower half of the feasible hierarchy. It retains at least n structural design degrees of freedom. It is a meaningful high-structural-accuracy regime, but it is not the maximal feasible range.

4. Minimax characterization

Write two structural responses as $r_i = p_i/q_i$, where q_i is monic of degree n .

Proposition 4 (Difference-numerator degree). *If $r_1, r_2 \in \mathcal{A}_{n,k}$ and $0 \leq k \leq 2n$, then*

$$r_1 - r_2 = \frac{H}{q_1 q_2}, \quad \deg H \leq 2n - k - 1,$$

unless the two responses are identical.

Proof. The generic numerator $p_1 q_2 - p_2 q_1$ has degree at most $2n - 1$. The shared Laurent coefficients imply $r_1 - r_2 = O(s^{-k-1})$. Since $q_1 q_2$ is monic of degree $2n$, the stated degree bound follows. $\square$

Define the relative structural error

$$\varepsilon_Q(s) = \frac{r_Q(s)}{F(s)} - 1.$$

Theorem 5 (Alternation certificate). *Let $0 \leq k < 2n$. Suppose $Q^* \in \mathcal{A}_{n,k}$ has $2n - k + 1$ ordered points on the design interval at which ε_{Q^*} has equal magnitude and alternating sign, and this magnitude equals its uniform norm. Then no rule in $\mathcal{A}_{n,k}$ has strictly smaller uniform relative error.*

Proof. A strictly better competitor would make $\varepsilon_{Q^*} - \varepsilon_Q$ alternate sign at the $2n - k + 1$ points, forcing at least $2n - k$ distinct zeros. Since $F(s) > 0$, these are zeros of $r_{Q^*} - r_Q$, contradicting the degree bound $2n - k - 1$, unless the responses are identical. $\square$

Remark 6. The theorem proves minimaxity within the stated class, not global uniqueness of the nonlinear parameterization. The zero-dimensional endpoint $k = 2n$ is determined by its moments and does not require a nontrivial alternation argument.

5. Numerical realization of the model-aware path

The constrained minimax equations combine the k moment conditions, equal-magnitude alternating errors, and stationarity at internal extrema. Lower-half rules were constructed in high precision. The missing upper-half problems were solved by continuation toward the Gaussian endpoint and refined through the equioscillation equations. All published rules have ordered nodes and positive weights.

Table 1: Complete $n = 4$ hierarchy. The endpoint $k = 8$ is zero-dimensional; therefore the alternation-certificate column is not applicable.

k	Dimension	Exactness	Cert. points	Worst relative error	Min. w_j
0	8	–	9	1.490623e-10	0.039147
1	7	1	8	2.460626e-09	0.044528
2	6	3	7	4.048641e-08	0.051670
3	5	5	6	6.653546e-07	0.061646
4	4	7	5	1.092819e-05	0.076595
5	3	9	4	1.794333e-04	0.101407
6	2	11	3	2.943219e-03	0.149011
7	1	13	2	4.597982e-02	0.126194
8	0	15	–	4.300499e-01	0.101229

For $n = 4$, the error grows from 1.49×10^{-10} at $k = 0$ to 4.30×10^{-1} at the Gauss endpoint. The upper-half values $k = 5, 6, 7$ continue the approximately constant finite-range price per additional moment before the error enters an order-one regime. This pattern is empirical and is not claimed as an asymptotic theorem.

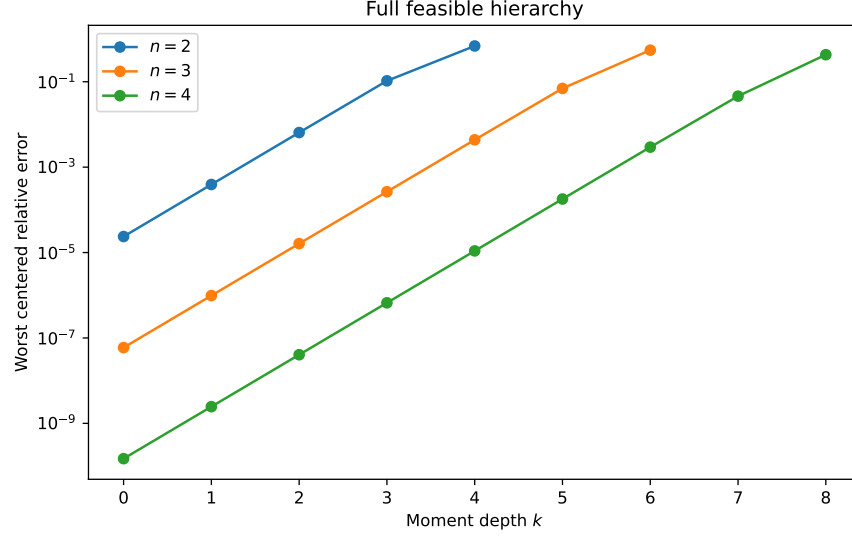


Figure 1: Worst centered relative error for the full feasible hierarchies $0 \leq k \leq 2n$, $n = 2, 3, 4$. The final point of each curve is the corresponding Gauss-Legendre rule.

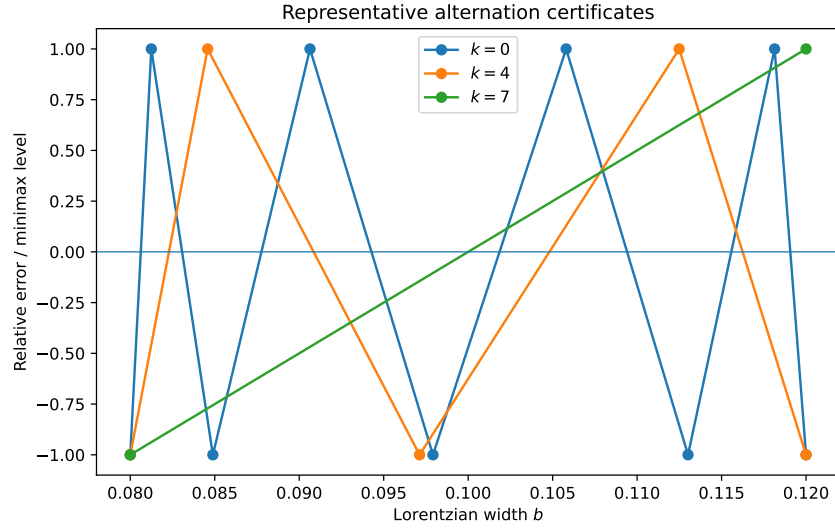


Figure 2: Representative normalized alternation patterns from the lower endpoint ($k = 0$), midpoint ($k = 4$), and upper half ($k = 7$) of the $n = 4$ hierarchy.

6. Exact coefficient-space rule selection

Let the uncertain smooth residual have a Chebyshev expansion

$$h(x) = \sum_{j=0}^{\infty} a_j T_j(x), \quad |a_j| \leq C \rho^{-j}, \quad \rho > 1.$$

For a fixed rule, define

$$\Delta_j(Q) = I[T_j] - Q[T_j].$$

Theorem 7 (Exact coefficient-box norm). *For every fixed bounded quadrature rule,*

$$\sup_h |I[h] - Q[h]| = CG_Q(\rho), \quad G_Q(\rho) = \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

Proof. Uniform convergence allows termwise application of $I - Q$. The triangle inequality gives the upper bound, and it is attained by the sign-aligned coefficients

$$a_j = C \rho^{-j} \operatorname{sign} \Delta_j(Q).$$

□

For the mixed class

$$f(x) = h(x) + cL_b(x), \quad |c| \leq c_{\max},$$

define

$$S_Q = \max_{b \in B} |I[L_b] - Q[L_b]|.$$

Theorem 8 (Exact mixed-class risk). *For every fixed rule,*

$$\sup_f |I[f] - Q[f]| = CG_Q(\rho) + c_{\max} S_Q.$$

The two signs can be chosen independently. For $C > 0$, define $\tau = c_{\max}/C$. The model-aware rule is therefore

$$k^*(\rho, \tau) \in \arg \min_{0 \leq k \leq 2n} [G_k(\rho) + \tau S_k].$$

Table 2: Full $n = 4$ structural and smooth-risk trade-off at $\rho = 1.5$.

k	Structural upper enclosure S_k	$G_k(1.5)$
0	5.55627e-09	9.11242e-01
1	9.17189e-08	3.95465e-01
2	1.50911e-06	2.04657e-01
3	2.48007e-05	9.59856e-02
4	4.07341e-04	4.49242e-02
5	6.68824e-03	2.14018e-02
6	1.09706e-01	1.17665e-02
7	1.71386e+00	6.62192e-03
8	1.60298e+01	3.42291e-03

7. Validated partition for the rounded rules

The exact hierarchy theory concerns ideal moment-constrained rules. Published decimal nodes and weights have small rounding defects. The coefficient-box risk remains exact for each fixed rounded operator, but a nominal moment label alone does not transfer exact symbolic properties.

We therefore re-evaluated all nine rounded $n = 4$ rules. The expected alternation counts were recovered for $k = 0, \dots, 7$. Structural errors were enclosed by a dense partition and inflated local derivative bounds. Chebyshev defects were evaluated through degree 500, and the remaining tail was bounded geometrically. A parameter cell was assigned a rule only when its risk upper bound was below the lower bounds of all competitors throughout the cell.

Adaptive refinement assigned a unique winner on

$$97.9067\%$$

of

$$(\rho, \log_{10} \tau) \in [1.05, 5] \times [-12, 12].$$

The unresolved area, 2.0933%, is concentrated near transition curves.

This validation is numerical rather than proof-assistant formal. The reported coverage depends on the stated enclosures and refinement depth.

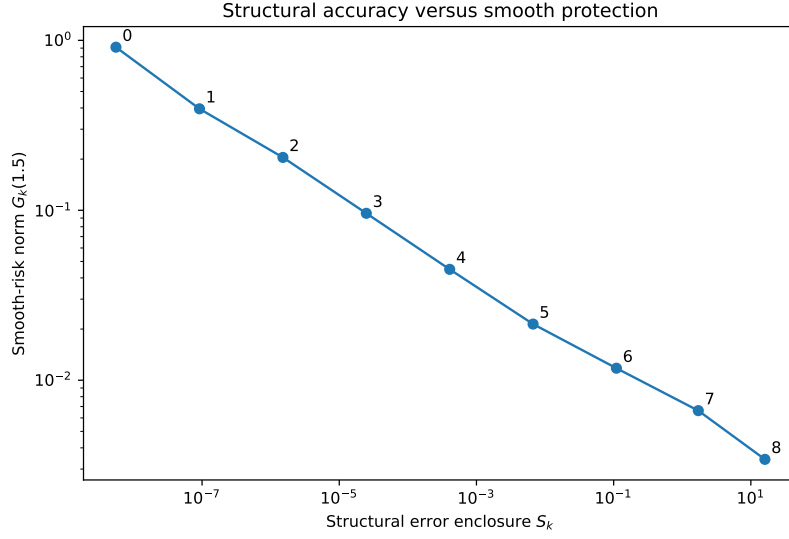


Figure 3: Each added moment decreases the smooth-risk norm but increases the Lorentzian structural error. Labels indicate the moment depth k .

Table 3: Fraction of the certified parameter area assigned to each moment depth.

k	Certified-area fraction
0	22.279%
1	9.216%
2	8.299%
3	8.463%
4	8.478%
5	8.460%
6	8.117%
7	6.934%
8	19.753%

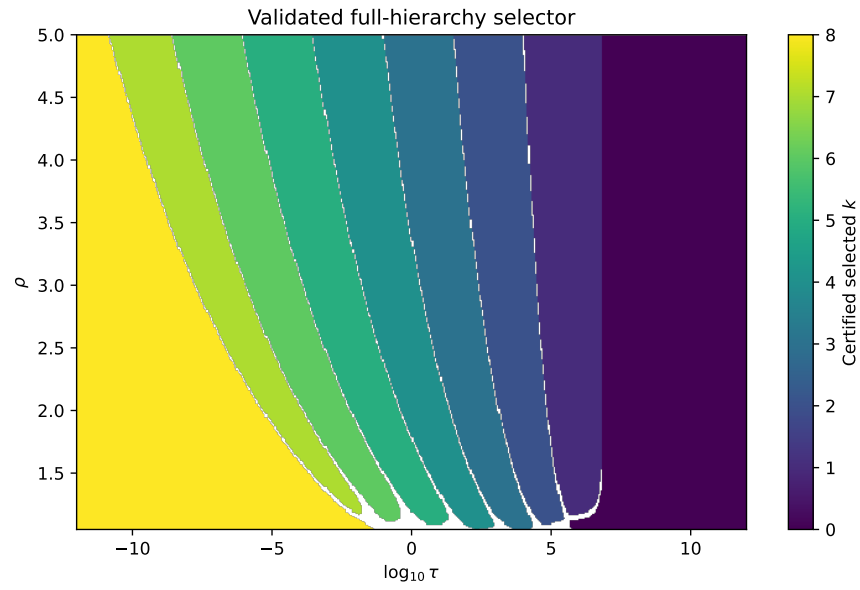


Figure 4: Validated full-hierarchy selector. Blank regions are not assigned because the conservative risk enclosures overlap.

8. External validity and reconstruction

8.1. Small pole-location mismatch

Without retraining, the centered rules were tested on

$$L_{a,b}(x) = \frac{1}{(x-a)^2 + b^2}, \quad |a| \leq 0.03, \quad 0.08 \leq b \leq 0.12.$$

Table 4: Worst relative errors on the shifted-Lorentzian stress-test box.

Method	Evaluations	Worst error	Worst $ a $	Worst b
PMC-k0	8	4.522336e-06	0.0300	0.0800
PMC-k4	8	1.025564e-03	0.0300	0.0845
Composite-Simpson-9	9	7.426762e-02	0.0300	0.1200
Gauss-Legendre-8	8	4.300499e-01	0.0000	0.0800

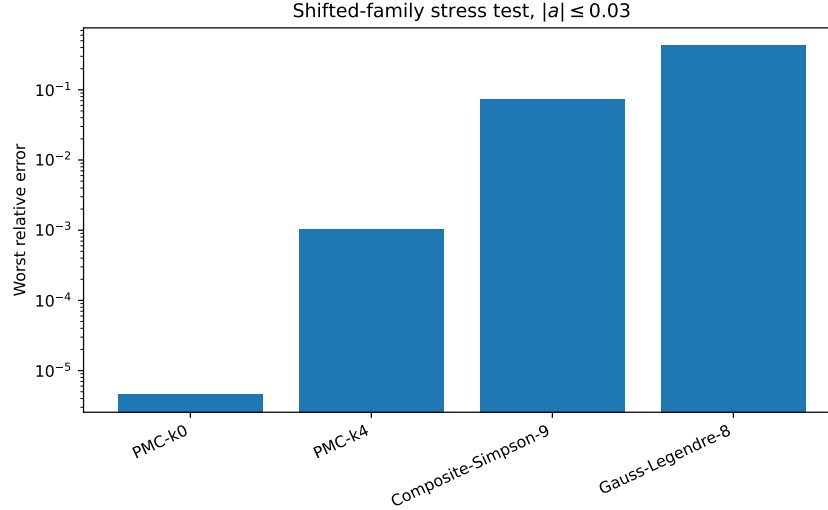


Figure 5: Stress test under small pole-location mismatch. The comparison supports local structural robustness, not shifted-family minimaxity or universal superiority.

The $k = 0$ rule has worst relative error 4.52×10^{-6} , whereas the Gauss endpoint has worst error 4.30×10^{-1} on the same box. The centered error is nevertheless inflated by more than four orders of magnitude, so the correct interpretation is local robustness rather than shift invariance.

8.2. Independent multi-start reconstruction

The six upper-half problems

$$(2, 3), (3, 4), (3, 5), (4, 5), (4, 6), (4, 7)$$

were reconstructed from twelve perturbed starts each. A run was accepted only when nodes were ordered, weights were positive, moment residuals were below 10^{-10} , and scaled equioscillation residuals were below 10^{-8} .

Table 5: Independent multi-start reconstruction of the upper-half rules. Error differences are relative to the published minimax level.

(n, k)	Accepted starts	Error difference	Max. node difference	Max. weight difference
(2, 3)	12/12	2.64e-16	5.55e-17	5.55e-17
(3, 4)	10/12	6.78e-14	4.44e-16	5.00e-16
(3, 5)	12/12	5.31e-14	1.29e-14	1.32e-14
(4, 5)	8/12	2.38e-12	4.44e-15	4.27e-15
(4, 6)	11/12	9.22e-14	1.03e-14	1.05e-14
(4, 7)	12/12	5.11e-13	8.95e-14	6.43e-14

All accepted solutions in each problem formed a single error-level cluster. No accepted start found a better rule by more than 10^{-6} relative, and the maximum best-rule error difference was 2.38×10^{-12} . This reduces the risk of a local numerical artifact but is not a proof of global uniqueness or third-party replication.

9. Relation to prior work and novelty scope

9.1. Structure-aware and generalized quadrature

Gautschi’s rational Gauss-type rules establish the principle that a quadrature formula can integrate rational functions selected to match important external poles [1]. The present work therefore does not introduce pole-aware quadrature. Its structural objective is different: it minimizes relative error over a continuum of Lorentzian widths instead of enforcing exactness at a finite set of prescribed poles.

Generalized Gaussian quadrature for Chebyshev systems supplies positive interior-node rules exact for non-polynomial spaces, and complete Chebyshev systems support stepwise increases in exactness [2]. Positive exact formulas and positive-node geometry are also well developed [6, 7]. Accordingly, neither positivity nor an ordered hierarchy of exactness is claimed as new. Here the

moment depth indexes a family selected by a continuum Stieltjes minimax criterion at fixed node-pair count.

Optimization-based nested quadrature shows that nonlinear moment-matching systems can be solved flexibly [8]. Our nonlinear solver and multi-start study are reproducibility mechanisms, not the principal conceptual contribution.

9.2. Rational approximation and minimax constraints

Rational minimax approximation and equioscillation are classical; adaptive barycentric algorithms make high-order problems computationally accessible [3]. Interpolation-constrained rational minimax approximation already includes prescribed function values and higher-order derivative conditions [4]. After the change of variable $z = 1/s$, matching Laurent coefficients at infinity is closely related to confluent interpolation at $z = 0$. Thus the novelty claim cannot be “minimax plus constraints.” The alternation theorem in this paper is instead a concise specialization to positive Stieltjes-response quadrature, where the shared coefficients reduce the difference-numerator degree.

Markov-function rational approximation provides especially relevant relative-error context [9]. Horning and Trefethen make the approximation-to-quadrature bridge explicit by deriving quadrature from rational approximations of Cauchy transforms; their framework also connects Gauss quadrature with Padé approximation at infinity [5]. This is the closest conceptual comparator. The distinction here is the simultaneous enforcement of positive residues, a variable polynomial-moment prefix, a continuum relative-minimax objective, and an exact mixed-class selector.

9.3. Worst-case rule selection

Worst-case analysis for analytic function classes is an established part of quadrature theory [10, 11], and recent positive-weight kernel quadrature gives a different convex-geometric worst-case framework [12]. The coefficient-box identity used here is elementary duality. Its role is to make the smooth residual commensurate with the Stieltjes structural error, producing

$$G_k(\rho) + \tau S_k.$$

The paper’s practically distinctive output is therefore not a new norm identity in isolation, but a selector and validated decision map over a finite family of positive structural rules.

9.4. *Defensible novelty statement*

The targeted audit found no direct source combining all of the following: positive symmetric Stieltjes residues, continuum Lorentzian relative minimax, variable moment depth at fixed node-pair count, an exact mixed smooth/structural selector, and a validated parameter partition. The defensible claim is this combined model-aware framework. The paper does not claim to invent positivity, moment matching, structure-aware quadrature, constrained rational minimax, equioscillation, or the Gauss/Padé connection separately.

10. Limitations

The theoretical and numerical study is one-dimensional and symmetric. The principal structural family is the centered Lorentzian kernel. The shifted-family experiment is a stress test and is not itself minimax-optimized or formally interval-certified. Global uniqueness of the nonlinear minimax parameters is not proved. The full selector validation leaves approximately 2.0933% of the domain unassigned. The empirical near-geometric price per moment is not an asymptotic theorem.

The literature audit is targeted rather than exhaustive. Equivalent constructions may exist under the language of moment spaces, generalized Gaussian quadrature, constrained Markov approximation, or interpolatory rational minimax. For this reason, the manuscript avoids “first” and “unprecedented” claims and retains specialist novelty review as a submission requirement. The phrase “validated numerical partition” refers to conservative computational enclosures, not proof-assistant verification.

11. Conclusion

Within the stated positive symmetric Stieltjes class, the variable moment depth organizes a path

$$k = 0 \quad \longrightarrow \quad k = 2n$$

from a structural minimax rule to Gauss–Legendre. The feasible-range characterization and difference-degree argument support the numerical construction, while the exact mixed risk turns the family into a model-aware selector. Complete computations for $n = 2, 3, 4$, conservative decision-map validation,

a shifted-pole stress test, and multi-start reconstruction quantify both the promise and the limits of the approach.

The main contribution is not any ingredient separately. It is the combination of positive Stieltjes realization, continuum structural minimax, variable polynomial protection, and exact rule selection in one computable framework. The next scientific step is independent specialist comparison with the closest generalized-quadrature and constrained-rational literature, followed by extensions beyond symmetry and the centered kernel.

Data and code availability

The code, published rules, generated data, and reproduction scripts are available at <https://github.com/TurkeyOp/partial-moment-stieltjes-quadrature>. The Version 2 upper-half and full-selector materials are included in the accompanying research archive and should be merged into the public repository before journal submission.

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During the preparation of this work, the author used OpenAI ChatGPT to assist with literature organization, code prototyping, manuscript structuring, and language revision. The author reviewed and verified the mathematical derivations, numerical results, references, interpretations, and final wording and takes full responsibility for the content of the article.

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